PROJECT PROPOSAL

CyberLab: Static Binary Analysis Toolkit

Static Binary Analysis Toolkit

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Department	Information and Communication Technology (ICT)
Course	Software Engineering / Software Project Management
Programming Language	Java
Interface	JavaFX
Database	MySQL
Proposed Duration	8 Weeks

Initial Proposal Draft

1. Abstract

CyberLab is a Java desktop application for basic static analysis of executable files without running them. It will identify file metadata, calculate SHA-256 hashes, extract printable strings, inspect available imported functions, flag rule-based indicators, and save/export analysis reports. The interface will use JavaFX and the database will be MySQL.

2. Problem Statement

Beginners in cybersecurity often need several tools to inspect executable files and separately record their findings. CyberLab aims to bring essential static-analysis tasks into one simple interface and provide organized analysis history and reports.

3. Objectives

• Identify supported ELF and PE files and show basic metadata.
• Calculate SHA-256 hashes and extract printable strings.
• Inspect imported functions where supported by the parser.
• Highlight potentially suspicious indicators using transparent rules.
• Save analysis history in MySQL and export reports as TXT or JSON.

4. Scope

The first version will support file selection, metadata inspection, hashing, strings extraction, basic import analysis, rule-based indicators, report export, and analysis history. It will not execute binaries or claim definitive malware detection. Advanced decompilation, sandboxing, and automatic exploitation are outside the initial scope.

5. Technology Stack

Programming language: Java. User interface: JavaFX. Database: MySQL. Database access: JDBC with a MySQL Connector/J driver. Build tool: Maven. Hashing: Java MessageDigest. Binary parsing: a suitable Java library or a limited custom parser. Diagrams: PlantUML or diagrams.net. Documentation: LaTeX.

6. Software Process Model

Agile development will be used to deliver the system incrementally. Each iteration will implement and test a small group of features, allowing issues to be identified early and the scope to be adjusted if necessary.

7. Schedule, Milestones and Deliverables

Week 1: Requirements and feasibility study — requirements specification.
Weeks 2–3: UML and database design — six diagrams and MySQL schema.
Weeks 4–5: Core analysis and JavaFX interface — working prototype.
Week 6: MySQL integration and report export — integrated application.
Week 7: Functional and integration testing — test report.
Week 8: Documentation and GitHub preparation — final report and source repository.

8. UML and System Diagrams

The final report will include: (1) Data Flow Diagram (DFD), (2) Use Case Diagram, (3) Activity Diagram, (4) Entity Relationship Diagram (ERD), (5) UML Class Diagram, and (6) UML Object Diagram. The ERD will include BinaryFile, AnalysisReport, and Indicator entities, with one-to-many relationships from BinaryFile to AnalysisReport and from AnalysisReport to Indicator.

9. Main Classes and Interface

The JavaFX prototype will use at least six classes: BinaryFile (file metadata and hash), FileAnalyzer (coordinates analysis), StringExtractor (extracts printable strings), ImportAnalyzer (inspects imported functions), AnalysisReport (holds findings and exports reports), and ReportRepository (saves and retrieves reports through MySQL). The interface will provide file selection, an analysis button, results panels, history, and report export.

10. Requirements, Testing and Risks

Functional requirements include file selection, metadata inspection, hashing, string extraction, import inspection, indicator display, saving/retrieving history, and report export. Non-functional requirements include usability, reliability, maintainability, and safe handling of untrusted files. Testing will cover individual classes, MySQL integration, invalid inputs, and supported sample binaries. Risks include unsupported formats, malformed files, large inputs, inaccurate indicators, and database connection errors; validation, size limits, exception handling, and clear limitations will mitigate them.

11. Expected Outcome

The expected outcome is a working JavaFX desktop prototype, a MySQL database for analysis records, exportable reports, six system diagrams, test documentation, and a GitHub repository. CyberLab is an educational tool; indicators are not definitive malware verdicts, and analyzed files will not be executed.

12. References

Oracle Java Documentation: https://docs.oracle.com/en/java/
OpenJFX: https://openjfx.io/
MySQL Documentation: https://dev.mysql.com/doc/
Apache Maven: https://maven.apache.org/guides/
NIST SP 800-86: https://csrc.nist.gov/pubs/sp/800/86/final

Appendix A. Eight-Week Gantt Chart

Activity	W1	W2	W3	W4	W5	W6	W7	W8
Requirements	■							
UML & DB design		■	■					
Core analyzer & UI				■	■			
Integration						■		
Testing							■	
Documentation & submission								■